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**ASSOCIATION OF PM_{2.5} WITH DIABETES, ASTHMA, AND HIGH BLOOD
PRESSURE INCIDENCE IN CANADA: A spatiotemporal analysis of the impacts
of the energy generation and fuel sales**

Article Post-Print

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Abstract

Numerous studies have reported an association between fine particulate matter (PM_{2.5}) and human health. Often these relationships are influenced by environmental factor that varies spatially and/or temporally. To our knowledge, there are no studies in Canada that have considered energy generation and fuel sales as PM_{2.5} effects modifiers. Determining exposure and disease-specific risk factors over space and time is crucial for disease prevention and control. In this study, we evaluated the association of PM_{2.5} with diabetes, asthma, and High Blood Pressure (HBP) incidence in Canada. Then we explored the impact of the energy generation and fuel sales on association changes. We fit an age-period-cohort as the study design, and we applied an over-dispersed Poisson regression model to estimate the risk. We conducted a sensitivity analysis to explore the impact of variation in clean energy rates and fuel sales on outcomes changes. The study included 117 health regions in Canada between 2007 and 2014. Our findings showed strong association of PM_{2.5} with diabetes, asthma, and HBP incidence. A two-year increase of 10 µg/m³ in PM_{2.5} was associated with an increased risk of 5.34% (95% CI: 2.28%; 12.53%) in diabetes incidence, 2.24% (95% CI: 0.93%; 5.38%) in asthma incidence, and 8.29% (95% CI: 3.44%; 19.98%) in HBP incidence. Our sensitivity analysis findings suggest higher risks of diabetes, asthma and HBP incidence when there is low clean energy generation. On the other hand, we found lower risk when we considered high rate of clean energy generation. For example, considering only diabetes incidence, we found that the risk in health regions with low rates of clean electricity is approximately 700%

higher than the risk in health regions with high rates of clean electricity. Furthermore, our analysis suggested that the risk in regions with low fuel sales is 66% lower than the risk is health regions with low rates of clean electricity. Our study provides support for the creation of effective environmental health public policies that take into account the risk factors present in Canadians health regions.

Keywords: Air pollution, Particulate matter, Diabetes incidence, Asthma incidence, High blood pressure incidence, Energy, Fuel.

1. INTRODUCTION

Several studies demonstrate that fine particulate matter (PM_{2.5}) exposure health effects. Worldwide, PM_{2.5} is responsible for 2.9 million deaths, according to the 2013 Global Burden of Disease Study (GBD 2015). The World Health Organization (WHO) estimates that PM_{2.5} is responsible for an 8.6 month reduction in life expectancy for the European population (WHO 2013). In China, Guo et al. (2016) show that the relative risks of lung cancer incidence related to a 10 µg/m³ increase in PM_{2.5} is 1.074 (95% CI: 1.060; 1.089). Pope et al. (2009) show that reductions in PM_{2.5} exposure during 1980s and 1990s led to an average increase in life expectancy of 4.9 months in United States. In Canada, studies have reported that long-term exposure to PM_{2.5} is associated with numerous diseases, which include lung cancer (Weichenthal et al. 2016), acute lower respiratory disease (Goldberg et al. 2013), myocardial infarction (Wang et al. 2015), diabetes (Goldberg et al. 2013; Brook et al. 2013), asthma (Villeneuve et al. 2007; Sahsuvargolu et al. 2009; To et al. 2013), and changes in blood pressure (Weichenthal, Hatzopoulou, and Goldberg 2014).

According to the Canadian Medical Association (CMA), in 2008, PM_{2.5} was linked to 92,000 emergency department visits (it is projected to reach 152,000 by 2031), and 21,000 early deaths (it is projected to be 39,000 in 2031) in Canada (CMA 2008). The total cost of air pollution for acute premature mortality between 2008 and 2031 in Canada is estimated to be over 250 billion dollars. Regionally these costs will be \$180 billion in the Central region, \$63 billion in the Western region, and \$16 billion in the Atlantic region (CMA 2008).

Determining the exposure and disease-specific risk factors over space and time is crucial for disease prevention and control. Several researchers have found that income, education, population density, neighborhood characteristics, traffic, and life style modify the effects of air pollution on human health (Requia, Roig, et al. 2016; Requia, Koutrakis, et al. 2016; Réquia Júnior, Roig, and Koutrakis 2015; Zou et al. 2014; Branis and Linhartova 2012; Rosenlund et al. 2008). For example, population density is related to poorer air quality (Hixson et al. 2012). Consequently, the regional distribution of costs of air pollution for premature mortality in Canada (mentioned above) is closely tied to the regional distribution of population (CMA 2008).

In this study, we evaluated the association of PM_{2.5} with diabetes, asthma, and High Blood Pressure (HBP) incidence in Canada. Then we explored the impact of the energy generation and fuel sales on association changes. Our hypothesis is that the

impacts from energy generation and fuel sales are significant environmental factors linked to the spatiotemporal association between PM_{2.5} and health. Although numerous studies have examined the risk factors related to the association between air quality and health, to our knowledge, there are no studies in Canada that have considered energy generation and fuel sales as modifier variables in a spatiotemporal analysis. Cities in Canada are experiencing growth that has been indicated to be increasing the vehicular emissions and the resulting ambient air pollution conditions (Adams et al. 2012). Our study can provide guidance to public policymakers when they invest in clean technologies, specifically providing locational awareness by health region where the greatest improvements to human health is possible.

2. MATERIALS AND METHODS

2.1. Study design and data

We conducted a cross-sectional analysis of the association between PM_{2.5} and diabetes, asthma, and HBP incidence with a focus on long-term PM_{2.5} exposures. The study was performed in five stages: i) health data processing; ii) covariates data processing; iii) exposure assessment – PM_{2.5} data; iv) statistical analysis, and; v) sensitivity analysis.

2.2. Health data processing

We used population-based Canadian community health survey data which were obtained from Statistics Canada (SC 2016). This dataset included the number of individuals diagnosed with diabetes, asthma or HBP by health region in Canada between 2007 and 2014. Health regions are administrative areas defined by provincial ministries of health according to provincial legislation. There are 117 health regions in Canada, which are presented in Figure 1 – panel A. The health data included detailed information on year, age, and gender. We limited our analysis to individuals above 44 years old, because there were fewer cases below this age. We considered 2 age groups: 45-64 years, and ≥65 years, which are consistent with the age categories in the Canadian census. Population data by age, gender and health regions were used to calculate the rate of health outcome incidence. These data were provided by the Statistics Canada (SC 2016).

2.3. Covariates data processing

The energy generation data were obtained from the DMTI Spatial (DMTI 2016), which included the geographic coordinates of power plants across Canada and the type of power plant. The power plant types in Canada included nuclear, hydro, coal, natural gas, diesel, fuel oil, and wind.

We processed the energy generation data using the following procedures. First, we stratified the type of power plants into two groups, which included clean electricity production (nuclear, hydro, wind), and dirty electricity production (coal, oil, natural gas, diesel, and fuel oil). Then we applied kernel density (GIS technique) to estimate the spatial density of each group of power plants. The kernel density is described as follows:

$$\lambda(u) = \frac{1}{\tau^2} \sum_{i=1}^n k\left(\frac{d(u_i; u)}{\tau}\right) \quad (1)$$

where λ is the estimator Kernel expressed in power plants points / area (km²); $u_1, u_2, \dots, u_n$ are the locations of n events observed based on the power plants locations; τ is the radius around u , which was 2 km; d is the distance between the position and i -ésima sample; and k is the function of estimation with the softening properties.

Finally, to obtain the level of clean energy generation, we calculated the rate: λ for clean electricity production / λ for dirty electricity production. Then we used a matching process to import the rate of clean energy to each health region.

Fuel sales data were obtained from Statistics Canada (SC 2016), which included detailed information on total net sales of gasoline in liter per year, by Canadian province. The fuel sales data were combined with the total road motor vehicle registrations by year and Canadian province to calculate the rate of fuel sales (net sales of gasoline per road motor vehicle). To estimate fuel sales by health region we applied a linear regression model with population as the predictor variable. The model was well fit with an R^2 of 0.92 ($p < 0.001$). The model was: Fuel sales = 75,050 + 1.139 population.

2.4. Exposure assessment

We assigned estimates of exposure to ambient PM_{2.5} derived from satellite remote sensing observations at 0.01 degree x 0.01 degree spatial resolution (van Donkelaar et al. 2015). We considered in our analysis the annual mean ambient concentration of PM_{2.5} for the period 2005 to 2014. The PM_{2.5} data were generated by combining Aerosol Optical Depth (AOD) retrievals with a chemical transport model, and then calibrated to ground

based observations of PM_{2.5} using Geographically Weighted Regression (GWR). For each health region we calculated the mean concentration within the health region boundaries. The estimates of PM_{2.5} do not cover the sparsely inhabited northern areas of Canada, which were excluded from our study. We used ArcGIS version 10.3 to process all spatial data.

2.5. Statistical analysis

Our statistical analysis was performed using an over-dispersed Poisson regression model (Berk and MacDonald 2008; Phang and Loh 2014). We fit an age-period-cohort model to separate the effects of age from the effect of risk factors in relation to year (Robertson and Boyle 1998). Several studies have shown significant evidence that incidence of diabetes, asthma, and HBP are linked to time-related variation (Harjutsalo, Sjoberg, and Tuomilehto 2008; Kwon et al. 2008; Brogger et al. 2004; Okamoto et al. 2007; Dobbin et al. 2004; Menotti et al. 2006). Therefore, first we included variables for age, period, and cohort effects; and then we extended the model with PM_{2.5} (including 2-year average) and other covariates. The model applied in our analysis is included in equation 2.

$$\begin{aligned} \text{Log}(\mu_{t,i,j}) = & \alpha + \beta_1 \text{PM}_{2.5} + \beta_2 \text{Age} + s(\text{period}) + \beta_3 \text{Cohort} \\ & + \beta_4 \text{Gender} + s(\text{spatial term}) + s(\text{energy}) + s(\text{fuel}) \\ & + \text{offset}(\log(\text{population})) \end{aligned} \quad (2)$$

where each data point is for subject j in the health region i ; t is the year of the observation; $\mu_{t,i}$ is the expected yearly incidence counts on the year t in the health region i ; α is the regression intercept; β values are regression coefficients for the categorical variables $\text{PM}_{2.5}$, Age (age group), Cohort (birth cohort – year of birth), and Gender ; period is the year of health outcome incidence; spatial term represents latitude and longitude, which was used to control for the spatial-autocorrelation of health outcome incidence; energy represents the rate of clean electricity production; and fuel is the rate of fuel sales.

The variables period , spatial term , energy , and fuel were modeled with natural cubic splines, denoted by $s(x)$, to characterize nonlinear relationships. We used Akaike information criterion for quasi-Poisson models (Q-QIC) to specify the degrees of freedom for each spline function. Finally, we included in the model the log of the population size

for each health region, for each gender, and for each age group as the offset. We used the software R - version 2.13.1 (R Core Team 2013) to perform all statistical analysis.

2.6. Sensitivity analysis

We conducted a sensitivity analysis to examine the robustness of the primary results, and to explore the impact variation of clean energy rates and fuel sales on outcome changes. We stratified our data in three different ways to perform the sensitivity analysis.

In the first stratification (sensitivity analysis 1), we performed cluster analysis in order to identify homogeneous groups of health regions based on the rate of clean energy generation and fuel sales variables, while maximized the heterogeneity between groups. We used the geostatistical method K-mean clustering in ArcGIS, version 10.3.

The K means algorithm identifies a seed feature randomly. Then all features are assigned to the closest seed feature. Finally, a mean data center is calculated for each group of features, and each feature is reassigned to the closest center.

We configured the K mean algorithm to group features using the K nearest neighbors approach. This approach determines that a health region will only be included in a group if at least one other health region in the same group is a natural neighbor.

The number of seed features selected randomly will match the number of groups. To estimate the optimal number of groups, we applied the Calinski-Harabasz pseudo F-statistic, which is described by the following equations:

$$CH = \frac{\left(\frac{T^2}{nc - 1} \right)}{\left(\frac{1 - T^2}{n - nc} \right)} \quad (3)$$

where CH is the Calinski-Harabasz pseudo F-statistic; nc is the number of classes (groups); n is the number of features; and T^2 is described as:

$$T^2 = \frac{SST - SSE}{SST} \quad (4)$$

where SST is a reflection of between-group differences (described by the Equation 5) and SSE is a reflection within-group similarity (described by the Equation 6).

$$SST = \sum_{m=1}^{nc} \sum_{j=1}^{ni} \sum_{k=1}^{nv} (V_{mj}^k - \bar{V}^k)^2 \quad (5)$$

$$SSE = \sum_{m=1}^{nc} \sum_{j=1}^{ni} \sum_{k=1}^{nv} (V_{mj}^k - \bar{V}_g^k)^2 \quad (6)$$

where ni is the number of features in group m ; nc is the number of classes (groups); nv is the number of variables used to group features (two variables - rate of clean energy generation and fuel sales); V_{mj}^k is the value of the k^{th} variable of the j^{th} feature in the m^{th} group; $\bar{V}^k$ is the mean value of the k^{th} variable; and $\bar{V}_g^k$ is the mean value of the k^{th} variable in group m .

In the second stratification (sensitivity analysis 2), we divided the health regions based on the level of clean energy generation and fuel sales (high/low values). For the low values we considered those below the 0.25 quantile, whereas for the high values we considered those greater than the 0.75 quantile. As a result, we generated four sub groups: i) high rate of clean energy generation, ii) low rate of clean energy generation, iii) high fuel sales, and; iv) low fuel sales.

In the third stratification (sensitivity analysis 3), we divided health regions using the same procedure from the sensitivity analysis 2. The difference here is that we considered the combination of the covariates. So we generated the following groups: i) high rate of clean energy generation and high fuel sales, ii) high rate of clean energy generation and low fuel sales, iii) low rate of clean energy generation and high fuel sales, and; iv) low rate of clean energy generation and low fuel sales.

3. RESULTS

Table 1 presents the number of diabetes, asthma, and HBP cases by gender and by age categories. There were a total of 5,570,326 diabetes cases, 2,592,161 asthma cases, and 17,541,801 HBP cases. Male cases of diabetes accounted for 55% of the cases reported. While females demonstrated a higher prevalence for cases for asthma (78%), and HBP (52%). The age group 45 – 64 had the highest number of cases for asthma incidence (65%), while the age group ≥ 65 had the highest number of cases for diabetes incidence (56%). For HBP incidence, the difference between the age groups is very small (Table 1).

Table 1 – Diabetes, asthma, and HBP cases in Canada, 2007 – 2014.

Group	Diabetes	Asthma	HBP	TOTAL
Male	3,074,708	555,075	8,403,427	12,033,210
Female	2,495,618	2,037,086	9,138,374	13,671,078
TOTAL	5,570,326	2,592,161	17,541,801	25,704,288
Age 45 – 64	2,402,887	1,707,066	8,756,019	12,865,972
Age ≥ 65	3,167,439	885,095	8,785,782	12,838,316
TOTAL	5,570,326	2,592,161	17,541,801	25,704,288

Figure 1 presents the spatial distribution of PM_{2.5}, type of energy generation, fuel sales, and health outcome incidence in Canada for the last year of our study period, 2014. The Southeast and Central-west are the regions with highest concentration of PM_{2.5}. Comparing maps (A) and (B.1) from the Figure 1, it is possible to see that the spatial distribution of PM_{2.5} is closely tied to the distribution of fuel sales.

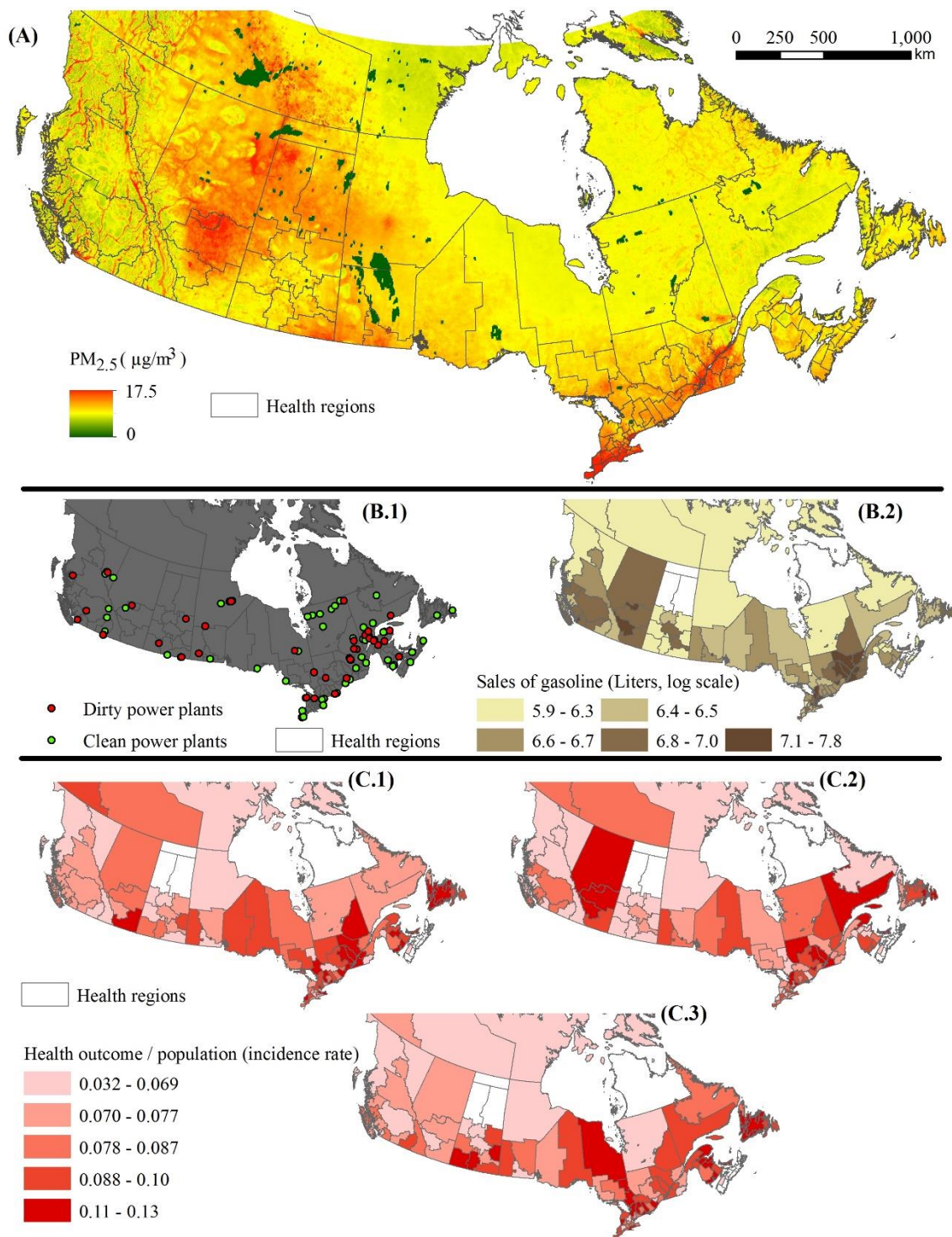


Figure 1 – Spatial distribution of $PM_{2.5}$, type of energy generation, fuel sales, and health outcome incidence in Canada in 2014.

Notes 1: Spatial distribution of $PM_{2.5}$ (A); Spatial distribution of energy generation (B.1); Spatial distribution of fuel sales (B.2); Spatial distribution of diabetes incidence (C.1); Spatial distribution of asthma incidence (C.2); Spatial distribution of HBP incidence (C.3).

Notes 2: The blank polygons of the maps B.2, C.1, C.2, and C.3 represent that we were unable to access data from those areas.

The associations between PM_{2.5} and diabetes, asthma and HBP incidence by different groups is presented in Table 2. A two-year increase of 10 µg/m³ in PM_{2.5} was associated with an increased risk of 5.34% (95% CI: 2.28%; 12.53%) in diabetes incidence, 2.24% (95% CI: 0.93%; 5.38%) in asthma incidence, and 8.29% (95% CI: 3.44%; 19.98%) in HBP incidence. Men had the highest risk in HBP incidence, while women had the highest risk in diabetes and asthma incidence. The age group 45 – 64 presented the highest risk in all health outcome incidence.

Table 2 – Percent increase (95% CI) in diabetes, asthma and high blood pressure incidence associated with an increase of 10 µg/m³ in PM_{2.5} in difference groups in Canada, during 2007 – 2014.

Group	Diabetes	Asthma	HBP
Overall	5.34 (2.28; 12.53)	2.24 (0.93; 5.38)	8.29 (3.44; 19.98)
Male	4.84 (1.41; 16.56)	1.28 (0.24; 6.89)	13.01 (3.96; 34.62)
Female	5.66 (1.73; 18.52)	2.90 (1.03; 8.11)	4.88 (1.30; 18.25)
Age 45 – 64	6.15 (1.41; 19.73)	3.97 (0.75; 20.99)	11.09 (3.51; 34.98)
Age ≥ 65	5.06 (1.78; 14.40)	2.48 (0.41; 14.76)	5.99 (1.27; 28.13)

For the sensitivity analysis 1 (K mean analysis), we performed a pseudo F-statistic test to estimate the optimum number of groups for analysis that maximizes both within-group homogeneity and between-group heterogeneity. The F-statistic test indicated three groups to be optimal. In addition, for the K mean analysis, an R² value was calculated for each variable. The highest R² indicates the variable that separates the study area into groups most effectively. Therefore, specifically for our analysis, the higher the R² value, the better the discrimination among the health regions (R² value varies between 0 and 1). We found R² equal to 0.85 for rate of clean energy generation, and 0.61 for fuel sales.

Figure 2 shows the summary statistics (parallel box plot) for each variable in each group. Group 1 (blue) presented lower values (first quartile) for rate of clean energy generation, and median values for fuel sales. Group 2 (red) had approximately median values for rate of clean energy generation, and very high values (outlier values) for fuel sales. Group 3 (green) presented higher values (> third quartile) for rate of clean energy generation, and median values for fuel sales.

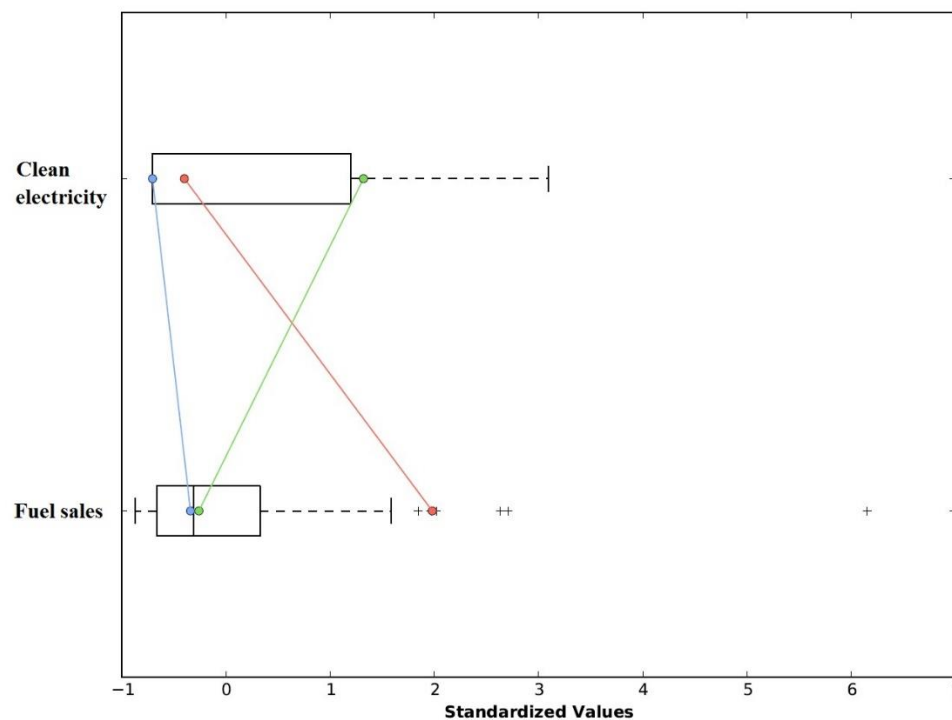


Figure 2 – K mean analysis – summary statistics (parallel box plot).
Notes: Group 1 (blue), Group 2 (red), Group 3 (green).

Figure 3 shows the risk for the primary and sensitivity analysis. For the sensitivity analysis 1, we present the risk only from the groups 1 and 3 (we did not find significant risk from the group 2). For the sensitivity analysis 2, we did not find significant risk from the high fuel sales group. So this group is not included in Figure 3. For the sensitivity analysis 3, we did not find significant risk. The number of cases in each strata for sensitivity analysis 3 was very low.

Our sensitivity analysis suggests higher risks of diabetes, asthma and HBP incidence when there is low clean energy generation. On the other hand, we found lower risk when we considered high rate of clean energy generation. For example, considering the diabetes incidence, our results showed a risk of 5.34% (95% CI: 2.28%, 12.53%) in the primary analysis, with a 17.25% (95% CI: 5.90%, 24.41%) risk in the sensitivity analysis 2 – low clean electricity, compared to a 2.07% (95% CI: 0.34%, 12.45%) in the sensitivity analysis 2 – high clean electricity. Comparing energy generation with fuel sales (e.g., low clean energy *versus* low fuel sales), our results showed lower risks for low fuel sales (Figure 3).

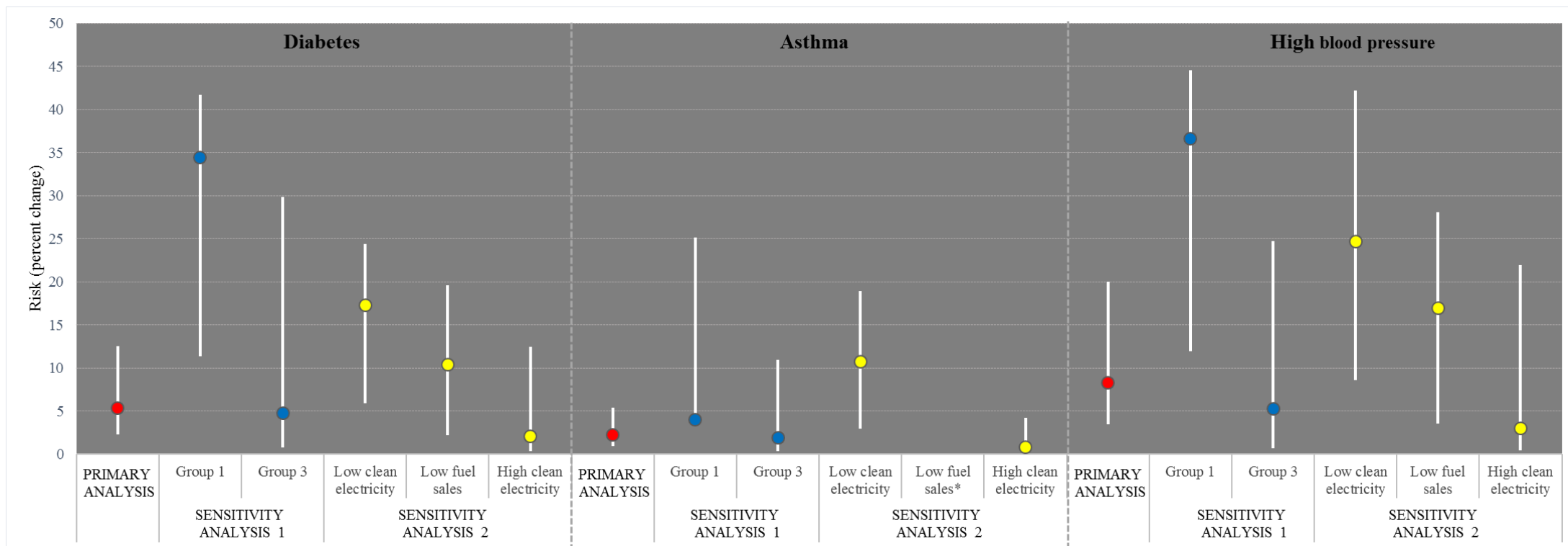


Figure 3 - Percent increase (95% CI) in diabetes, asthma and high blood pressure incidence associated with an increase of 10 $\mu\text{g}/\text{m}^3$ in $\text{PM}_{2.5}$ in Canada, during 2007 – 2014: primary analysis and sensitivity analysis.

Note: No significance, p-value > 0.05 (*)

4. DISCUSSION

We found a strong positive association of PM_{2.5} with diabetes, asthma, and HBP incidence, which is consistent with results from other studies in Canada and studies from other countries. An increase of 10 µg/m³ in PM_{2.5} was associated with an increased risk of 5.34% (95% CI: 2.28%; 12.53%) in diabetes incidence. Brook et al. (2013) used the same source for ambient PM_{2.5} that we applied (derived from satellite observations). They found that a 10 µg/m³ elevation in PM_{2.5} exposure was associated with an increase in risk for diabetes-related mortality in Canada, hazard ratio 1.49 (95% CI: 1.37; 1.62).

Fan et al. (2016) performed a meta-analysis on the association between asthma and PM_{2.5} exposure. The authors calculated the pooled risk ratio for 16 publications from USA, Canada, Finland and Taiwan, and found that a 10 µg/m³ elevation in PM_{2.5} was associated with a 1.5% (95% CI: 1.2%; 1.7%) increase in emergency department visits for asthma. In our study the increase in asthma incidence was 2.24% (95% CI: 0.93%; 5.38%). Cai et al. (2016) also performed a meta-analysis to assess the association between PM_{2.5} and HBP. They included 17 studies from 9 countries (including Canada) to estimate the pooled risk. Cai et al. (2016) found that long-term exposure to PM_{2.5} had positive relationships with HBP, but lacked statistical significance. They only found a significant association for short-term exposure, which the risk was 6.9% (95% CI: 0.3%; 14.1%). Our findings showed significant association between long-term exposure to PM_{2.5} and HBP, 8.29% (95% CI: 3.44%; 19.98%).

Our findings suggest that the rate of clean energy generation and fuel sales are strong modifiers of the association between PM_{2.5} and diabetes, asthma and HBP. For example, considering only diabetes incidence, we found that the risk for health regions with low rate of clean electricity is approximately 700% higher than that from health regions with high rates of clean electricity (sensitivity analysis 2), and 200% higher than that from the primary analysis. Comparing electricity with fuel sales, our results demonstrated that the risk from regions with low fuel sales is 66% lower than that from health regions with low rate of clean electricity (sensitivity analysis 2).

Other studies have also reported results suggesting the potential effects of energy generation and fuel sales on human exposure. In China, coal power plants are strongly associated to PM_{2.5}-related health impacts (Ji et al. 2015). In Santiago, Chile, diesel fuel consumption is a significant factor to understand PM_{2.5} trends and to inform air quality control efforts (Jhun et al. 2013). Razeghi et al. (2016) assessed the impacts of power generation, transportation on air quality in the USA, and found that PM_{2.5} decreased as a

result of the introduction of wind energy to the grid mix and the use of plug-in electric vehicles. Tessum et al. (2014) evaluated air quality (PM_{2.5} and O₃) related to human health impacts of fuels type including the use of electric vehicles, and use of electricity from a range of conventional and renewable sources in USA. The authors found that the health impacts vary by 80% depending on the use of fuel and electricity source. Electric vehicles powered by clean electricity (i.e., wind, hydropower, or solar power) reduce health impacts by at least 50%.

The impacts of energy generation and mobility on human health have been presented as a crucial concern worldwide. Electric vehicles have been considered as an alternative to minimize the health impacts related to internal combustion engines. Therefore, our study is useful and important, especially considering that several countries (including Canada) have initiatives that promote adoption of electric vehicles. Millions of electric vehicles have been sold around the world (Pike Research 2012). In 2009, the U.S. government has allocated considerable investments to promote the adoption of electric vehicles (Lieven et al. 2011). In Canada, approximately 2,000 electric vehicles were sold in 2012. An average sale of 107,000 electric vehicles per year has been predicted for the period 2015 and 2020 (Pike Research 2012).

The strengths of this current study include that our results identify a portion of the relationship between health and energy/mobility in Canada. Second, to our knowledge, there are no studies in Canada that have considered energy generation and fuel sales as modifier variables in a spatiotemporal analysis of the association between PM_{2.5} and diabetes, asthma, and HBP. Third, we used an age-period-cohort models to improve our estimates.

Some of the limitations to our current work include not controlling for individual data (e.g., smoke, alcohol, body mass index), since we were unable to access this data. This may bias the health effects estimates. It was necessary to estimate fuel sales for each region as these data were only available regionally. We used health data at the community level, but not at the individual level, which may also underestimate the risk (Borgini et al. 2011; Gerharz et al. 2013).

5. CONCLUSIONS

The rate of clean energy generation and fuel sales are strong modifiers to the association of PM_{2.5} with diabetes, asthma, and HBP incidence. This suggest that dirty energy generation (e.g. coal), and gasoline combustion from traffic may be damaging

human health. Our study provides support for the creation of effective environmental health public policies that take into account the risk factors present in Canadians health regions. In addition, the obtained results can be used as a guide for future policies, particularly the spatial regions that will have the greatest impact on human health improvement, in new technologies related to energy and mobility (e.g., electric vehicles). Finally, these results may guide policy makers, private sector, environmental and transportation agencies to develop strategies related to electric vehicles adoption in Canada and other countries.

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